

AI-100 Final project

What I Learned About GenAI

In this project, I learned that GenAI can act as a cognitive partner rather than just a tool that gives answers. Instead of directly solving problems, it helped me improve my reasoning and debugging process.

For example, when I used the wrong file name like sales.csv instead of summer_sales_dataset.csv, GenAI helped me realize that the file path might be wrong. Also, when I used wrong column names or added missing values, it guided me to check the dataset properly.

Another example is when I changed numeric values to text like price = "cheap". GenAI helped me understand that machine learning models need numbers, not text, and that data types are very important.

What I Learned About Building an AI System

In this project, I learned how an AI system works and how it is trained.

First, I understood that AI takes input data (features) and learns patterns from it. In my project, the model used features like price and other values to predict the product. The algorithm (Logistic Regression) learns from the data and then gives predictions.

I also learned that data is very important:

- If data has missing values, the model may not work.
- If data has wrong types (text instead of numbers), it gives errors.
- If data is incorrect, predictions will also be wrong.

Another important thing I learned is that small mistakes can break the system. For example:

- Predicting before training gives an error.
- Using wrong columns gives errors.
- Shuffling data incorrectly gives wrong results.

I also understood that AI works with numbers. Even things like images are converted into numbers before the model can learn from them.